Kriging in Sports

CMDA 4654 Project 2 - Group 1

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April 2026

Data Introduction

Points per shot is one of the biggest current topics of discussion in basketball analytics. It is calculated using two factors: a team or players chance of making a shot from a position on the floor and the point of value of a make from that position. Performing analysis on this topic requires gathering both of those values which can be done using the `hoopr` package.

```r
library(hoopR)

pbp <- load_nba_pbp(seasons = 2026)
player_box <- load_nba_player_box(seasons = 2026)

player_names <- player_box %>%
  distinct(athlete_id, athlete_display_name)

shots <- pbp %>%
  filter(shooting_play == TRUE,
         !is.na(coordinate_x), !is.na(coordinate_y), !grepl("Free Throw", type_text),
         season_type == 2) %>%
  select(athlete_id_1, coordinate_x, coordinate_y, score_value) %>%
  left_join(player_names, by = c("athlete_id_1" = "athlete_id")) %>%
  mutate(
    x = ifelse(coordinate_x > 0,
               -(coordinate_x - 46.75),
               coordinate_x + 46.75),
    y = ifelse(coordinate_x > 0, -coordinate_y, coordinate_y)) %>%
  filter(x <= 45) %>%
  select(athlete_display_name, score_value, x, y)
```

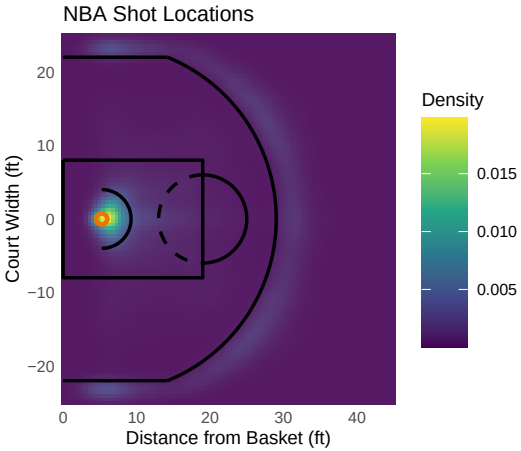
Data Example

The data has been loaded and saved as csv for ease of access: Click here to access shots.csv.

Preview of the data

athlete_display_name	team_name	score_value	x	y
Luka Garza	Boston Celtics	3	28	12
Franz Wagner	Orlando Magic	3	30	13
Luka Garza	Boston Celtics	0	24	17
Paolo Banchero	Orlando Magic	2	7	1
Baylor Scheierman	Boston Celtics	3	38	-1
Jalen Suggs	Orlando Magic	0	17	23
Ron Harper Jr.	Boston Celtics	0	27	15
Franz Wagner	Orlando Magic	3	4	-23
Baylor Scheierman	Boston Celtics	0	31	-8
Wendell Carter Jr.	Orlando Magic	0	15	23
Baylor Scheierman	Boston Celtics	0	8	23
Paolo Banchero	Orlando Magic	0	6	-3
Paolo Banchero	Orlando Magic	0	8	2
Paolo Banchero	Orlando Magic	0	5	0
Max Shulga	Boston Celtics	0	15	-2
Jordan Walsh	Boston Celtics	0	23	-21
Baylor Scheierman	Boston Celtics	0	10	-2
Ron Harper Jr.	Boston Celtics	2	8	-4
Ron Harper Jr.	Boston Celtics	2	7	-4
Desmond Bane	Orlando Magic	0	34	12

NBA 2025-26 Regular Season Heatmap



Kriging Motivation

It is clear from the shot density layout on the previous slide that there are two main areas that teams take shots from:

- the three point line
- the rim

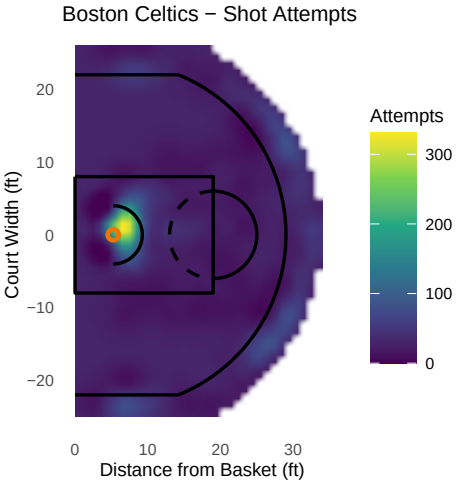
This makes sense given that shots at the rim are likely to be high percentage and shots from three point range are worth more points. The middle area between these two hot zones is fairly empty, and the league has been trending to make midrange jumpshots even less a part of offenses in the Modern Era. All of this evolution is in the pursuit of offensive efficiency which is where Kriging comes in.

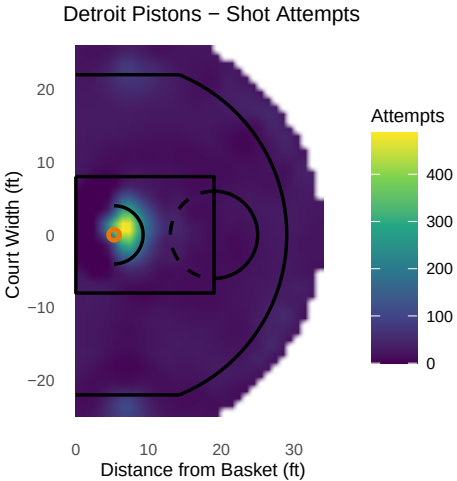
Since the shot map is made up of x and y data, it can be treated similarly to a geostatistics field from more typical Kriging - it has high density areas and low gaps with a large number of very individualized spots. Kriging can be used to smooth over these less commonly attempted areas and give a picture of not only where teams shoot but how well they perform from these areas.

Combining those two areas will mean that Kriging can make a points per shot estimation from every location on the floor which can then be used to check if teams are running their offenses to peak efficiency.

Attempt Frequency Kriging

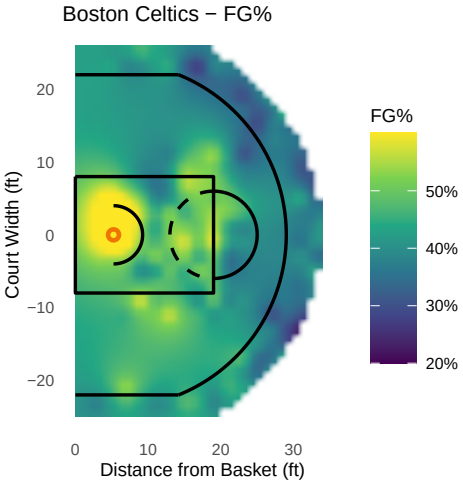
A typical picture of attempt frequency for a basketball team would consist of colored boxes or hexes that represent volume from each area of the court, but a custom Kriging can smooth this out into a continuous map using two foot by two foot bins.

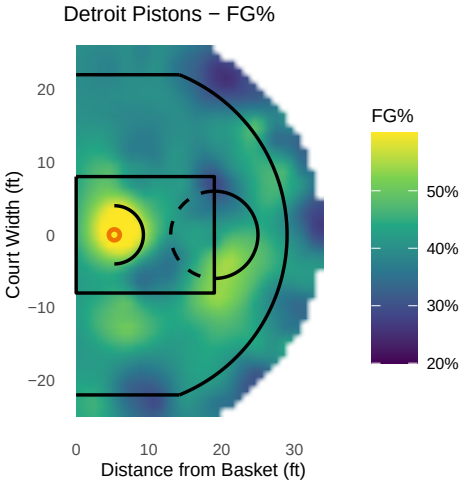




FG% Kriging

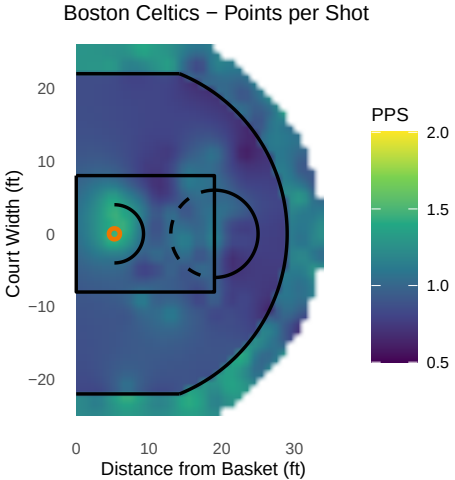
While the Kriging for shot attempts is somewhat trivial, it is extremely useful for field goal percentage. This statistic leads to a map that has continuous values across all the bins and the maps will differ more greatly team-to-team.

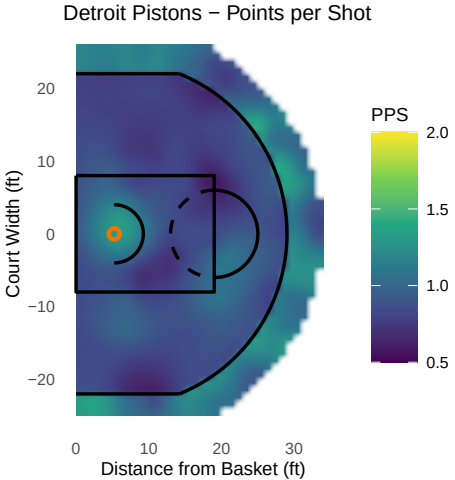




Points Per Shot

As mentioned previously, points per shot is a metric that is used to determine the quality of a type of shot attempt. Now that the field goal percentage Kriging has supplied a likelihood of making a shot at any location on the floor, it is easy to take it a step further into point values of the shots.



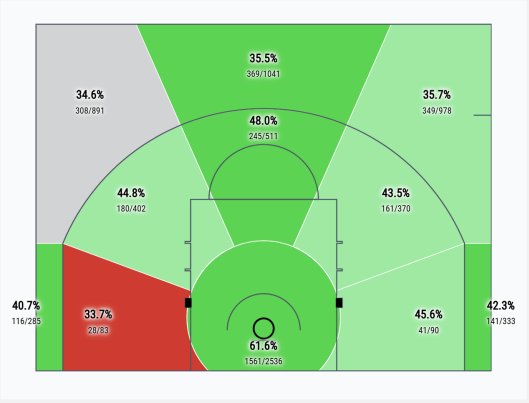


Non-Kriging Approach

So the question becomes "How do these smooth maps help NBA teams improve their offenses?"

One of the main advantages that these plots provide is the way they make it much easier to full picture of slots across the floor. As a comparison, here would be a more standard look at a teams percentages and shot volumes from Synergy Sports.

The plot makes it hard to tell whether there's smaller deviations within the large group and doesn't calculate the point values of the shots (i.e. shooting 50% from 2 is the same as 33% from 3). Thus it is generally not very insightful about which shots are the most efficient, only offering general statistics about which areas the team has shot the most accurately from.



Kriging Comparison

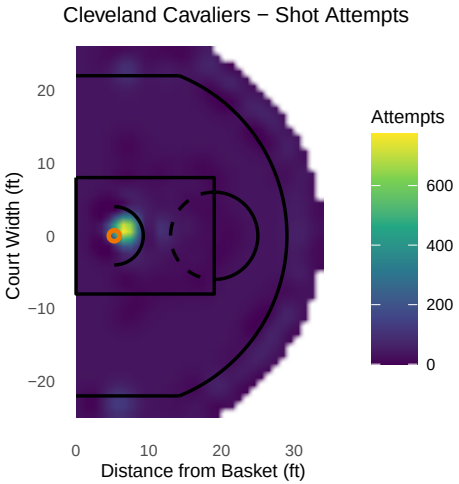
A smooth chart from Kriging much better captures the specific peaks and valleys of the shooting percentage and factoring in the point values gives real, actionable insights. Combining this with the shots attempted map can make it easy for a coach without a statistical background to see mismatches in shot value and frequency without complex interpretation.

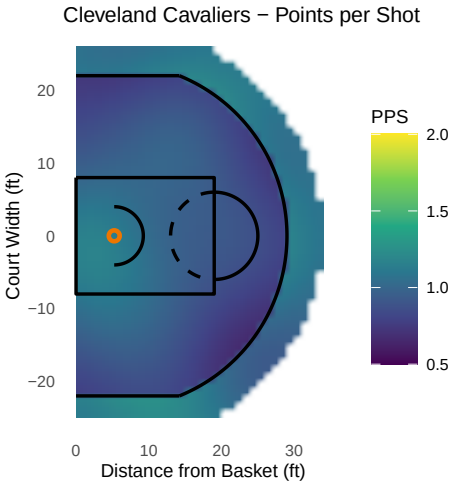
Running through every NBA team can make some interesting comparisons:

```
plot_team_impact <- function(team_name) {
   fg_obj <- plot_fg_pct(team_name)
   attempts_plot <- plot_attempts(team_name)
   pps_plot <- plot_pps(fg_obj$data)
   attempts_plot | pps_plot
}

nba_teams <- sort(unique(team_shots$team_name))

for (tm in nba_teams) {
   print(plot_team_impact(tm))
}
```

Team Takeaways

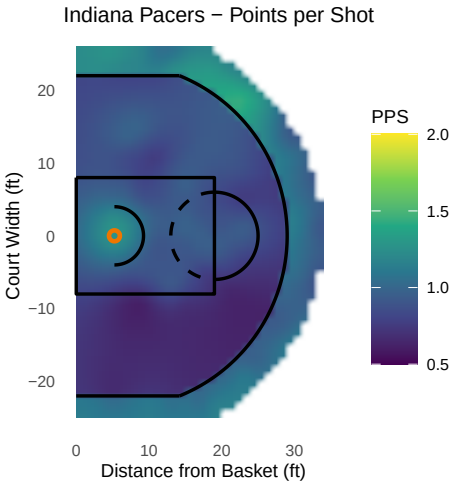
Some teams could gain insights by looking at their charts side-by-side, like the Cleveland Cavaliers

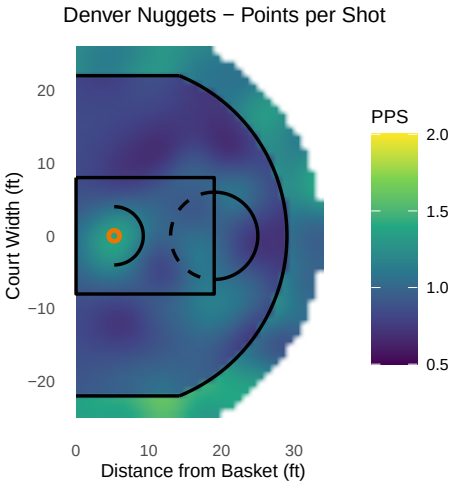




Points per Shot Patterns

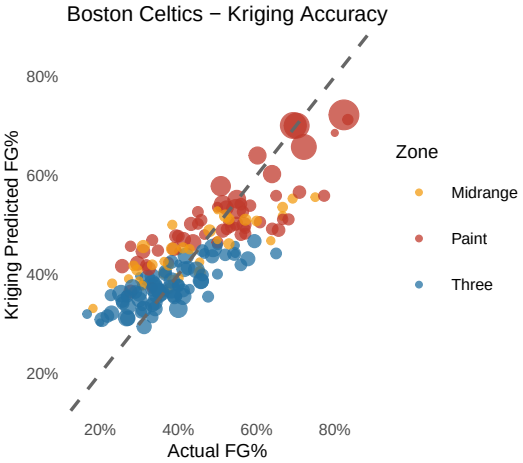
Some teams also have clear patterns that favor shooting from one side of the floor

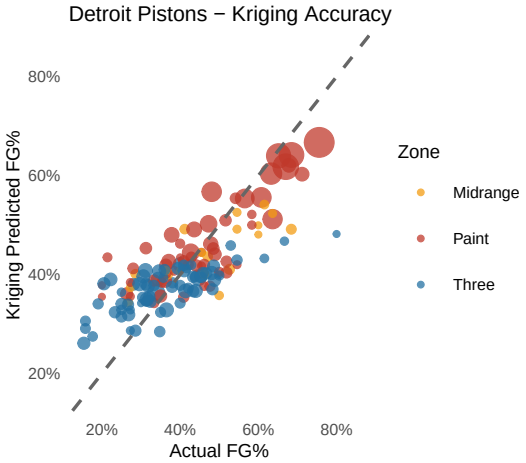




Accuracy of Predictions

Of course, the predicted maps are only useful if they are accurate. So how well did Kriging do with smoothing the shooting data?

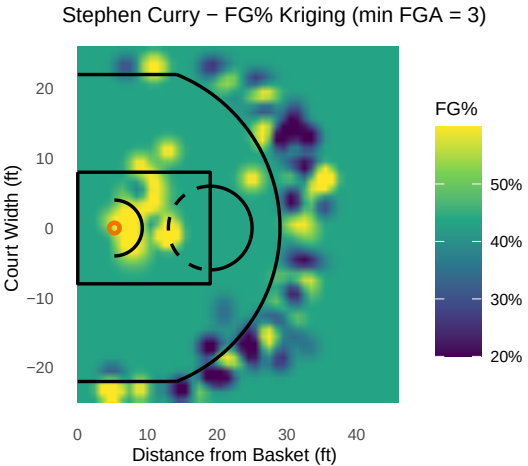




Kriging Limitations

Kriging has 2 major limitations: sparse data and giant data. The giant data problem is avoided for this case by filtering out all of the free throws and plays that did not result in a shot attempt from the large play-by-play dataset, but slight manual adjustments had to be made for sparse data:

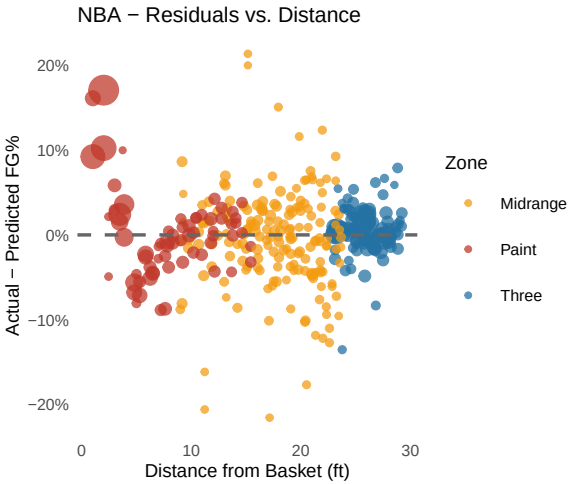
This plot of Stephen Curry's individual FG% Kriging from last season makes it clear that having too little data can be a problem as well. Not only are the bins that have enough shots to analyze somewhat blocky, but the areas where there are no data have been pulled towards the global mean which is obviously unreasonable given the context. This emphasizes that Kriging is far better at interpolation than extrapolation.



Conclusions

A plot of the residuals from the predictions against the distance from the basket of the shots gives some insights about how Kriging outputs tend to work - the model underpredicted the almost outlier-like high percentages on dunks and layups near the basket, was most wild for the midrange where the data is far more sparse and inconsistent, and then was fairly tight and accurate for three point shots.

This means that NBA teams could have the most confidence in conclusions related to threes but should perhaps be careful when making broad conclusions about midrange shots based on the Kriging technique.



References

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